What is it ?

- Complete Tree/Binary Tree
- children or grand children smaller than current node → Max heap → Root is largest
- children or grand children larger than current node → Min heap → Root is smallest

How does it work ?

- Work like a array under the hood: [0, 1, 2... n-1] with length n
- Children of node i: 2i+1 & 2i+2
- Parent of node i: [(i-1)/2]
- When add: insert at the end, bubbling up by comparing and swapping with its parent until the heap condition is satisfied
- When delete (remove the root): remove the first (head) element, then replace it with final node, then bubbling down
- Running time: O(LogN)

What can it do ?

- It is self balancing
- It can be used for priority (easily get the smallest or largest)